An AI-Driven Decision Support System for Tomato Cultivation in srilanka

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Tomato farming is one of the most economically important agricultural activities in Sri Lanka and across the world. However, many small-scale farmers face significant challenges in determining whether their soil conditions are suitable for tomato cultivation and what corrective actions to take when they are not. Existing IoT-based precision agriculture systems focus primarily on crop recommendation, yield prediction, or disease detection, but none provide a complete cycle of soil suitability diagnosis, corrective recommendation, and follow-up verification .

This research identifies a clear gap in the existing literature: while systems such as those proposed by Babu et al. , Islam et al. , Tifanto et al. , Saha et al. , and Cheema & Pires collect soil data and make predictions, they do not explain why the soil is unsuitable in a way the farmer can act on, they do not suggest practical corrective actions, and they do not track whether those corrections are working over time.

This project proposes an IoT-based smart soil health assessment system specifically designed for tomato farming. The system collects real-time soil data including nitrogen (N), phosphorus (P), potassium (K), pH, temperature, humidity, and soil moisture using low-cost sensors connected to an ESP32 microcontroller . A Random Forest machine learning model then classifies the soil as suitable or not suitable for tomato cultivation . When the soil is found to be unsuitable, SHAP (SHapley Additive exPlanations), an XAI (Explainable Artificial Intelligence) technology, identifies exactly which soil parameters are responsible for the problem . A rule-based recommendation engine then provides the farmer with specific corrective actions such as adjusting pH or applying fertilizer . The system also performs time-series tracking, comparing soil readings from two weeks ago, one week ago, and today, to verify whether the applied corrections are improving soil conditions . If insufficient improvement is detected, a revised recommendation is provided.

The system is designed to be practical, low-cost, and accessible to tomato farmers in Sri Lanka. The expected outcomes include a working prototype with over 95% prediction accuracy, a mobile-accessible dashboard, and a complete soil health feedback loop that no existing system in the literature currently provides, addressing the gap identified across reviewed studies . This system contributes to food security and SDG 2 (Zero Hunger) by empowering small-scale farmers with intelligent, data-driven soil management tools.

Keywords: IoT, Soil Health Assessment, Random Forest, SHAP, Explainable AI

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
IoT	Internet of Things
ML	Machine Learning
RF	Random Forest
SHAP	SHapley Additive exPlanations
NPK	Nitrogen, Phosphorus, Potassium
XAI	Explainable Artificial Intelligence
DHT11	Digital Humidity and Temperature Sensor
ESP32	Espressif Systems 32-bit Microcontroller
MQTT	Message Queuing Telemetry Transport
RAG	Retrieval Augmented Generation
DSS	Decision Support System
XAI	Explainable Artificial Intelligence
AI	Artificial Intelligence
SDG	Sustainable Development Goal
API	Application Programming Interface

1. INTRODUCTION

1.1 Background and Context

Tomato (*Solanum lycopersicum* L.) is one of the most widely cultivated vegetable crops in the world, with global production exceeding some million tons annually. In Sri Lanka, tomato farming is a primary source of income for a large number of smallholder farmers. However, these farmers frequently encounter poor yields due to unsuitable soil conditions, improper nutrient management, and lack of timely guidance on how to correct identified problems.

Soil health is a fundamental determinant of crop performance. Key soil parameters such as nitrogen (N), phosphorus (P), potassium (K), pH, temperature, moisture, and electrical conductivity directly affect tomato growth, disease resistance, and yield quality. For example, a soil pH below 6.0 can cause calcium and magnesium deficiencies in tomatoes, while excess nitrogen increases the risk of late blight disease. Suboptimal soil moisture, particularly below 24% volumetric water content, has been shown to suppress fruit expansion significantly. Studies confirm that soil temperature around 21.8°C and electrical conductivity between 0.6–0.8 dS/m are critical for optimal tomato yield. In Sri Lanka, smallholder farmers account for over 80% of total agricultural production, yet most lack access to affordable real-time soil monitoring and assessment tools.

Recent advances in IoT and machine learning have produced several promising precision agriculture systems. IoT-based soil monitoring systems such as those proposed by Postolache et al. and Correa-Quiroz et al. successfully collect real-time soil data but focus only on irrigation control without providing soil suitability assessment. Crop recommendation systems developed by Prity et al., Afzal et al., Cheema and Pires, and Ghosh et al. use Random Forest and other machine learning models to recommend suitable crops with 97–99% accuracy, but they do not assess whether existing soil is suitable for a specific crop already chosen by the farmer. Li et al. demonstrated the use of SHAP technology to explain soil parameter importance for tomato growth, but their system stops at explanation without connecting to actionable recommendations. Folorunso et al. and Islam et al. provide fertilizer recommendations based on soil data, but these recommendations are static and never verified against actual soil improvement over time. Martelli et al. and Saxena et al. collect time-series sensor data for irrigation management, but no existing study uses time-series soil data to verify whether a corrective intervention is working.

Despite the availability of IoT sensor technology and machine learning tools, the majority of existing precision agriculture systems focus only on prediction or classification. They do not guide the farmer through the full cycle of identifying a problem, fixing it, and verifying that the fix has worked. This creates a critical practical gap between technology and farm-level impact, particularly for resource-limited farmers who need clear, actionable guidance.

To address these limitations, this research proposes an IoT-based Smart Soil Health Assessment System for Tomato Farming consisting of five key components. First, an ESP32 microcontroller connected to NPK, pH, moisture, temperature, and humidity sensors collects real-time soil data. Second, a Random Forest machine learning model classifies the soil as suitable or not suitable for tomato cultivation. Third, SHAP (SHapley Additive exPlanations), an XAI technology, identifies which specific soil parameters are responsible for the unsuitable prediction. Fourth, a rule-based recommendation engine provides the farmer with specific corrective actions such as adjusting pH or applying fertilizer. Fifth, a time-series

tracking module compares soil readings from two weeks ago, one week ago, and today to verify whether the applied corrections are improving soil conditions, providing revised recommendations if improvement is insufficient.

Table 1:Optimal Soil Parameter Ranges for Tomato Cultivation

Parameter	Optimal Range	Source
Soil pH	6.0 - 6.8	Babu et al. [1], Postolache et al. [5]
Soil Temperature	18°C - 24°C	Martelli et al. [4], Li et al. [8]
Soil Moisture (VWC)	24% - 28.5%	Martelli et al. [4]
Electrical Conductivity	0.6 - 0.8 dS/m	Li et al. [8]
Air Humidity	60% - 70%	Saxena et al. [9]
Nitrogen (N)	Moderate - Not too high	Babu et al. [1]
Phosphorus (P)	Adequate level	Afzal et al. [7]
Potassium (K)	Adequate level	Afzal et al. [7]

1.2 Literature Review

A substantial body of recent literature has explored the use of IoT and machine learning in precision agriculture. The following review examines the most relevant studies and highlights their contributions and limitations.

Babu et al. [1] developed a hybrid ML framework integrated with IoT, using Arduino microcontrollers with NPK, temperature, humidity, and GPS sensors, deployed in tomato fields in Anakapalle, Andhra Pradesh, India. Their approach combining Bayesian Optimization with KNN achieved 95% accuracy in predicting tomato diseases including early blight, late blight, and Cercospora leaf mold. However, the system does not recommend corrective actions or monitor whether soil conditions improve over time.

Islam et al. [2] proposed an IoT system using NodeMCU (ESP8266) integrated with JXBS-3001 NPK sensors, FC-28 moisture sensor, and DHT11 temperature and humidity sensor, combined with CatBoost for crop recommendation (97.5% accuracy) and Random Forest with Grid SearchCV for fertilizer prediction (99.56% accuracy). Their system includes a cloud-hosted mobile application with QR-based crop history tracking for consumers. However, the system does not perform soil suitability diagnosis nor provide a feedback mechanism to verify whether the recommended actions have led to measurable improvement in soil conditions.

Tifanto et al. [3] designed a soil nutrient analysis and crop recommendation system using ESP32 and a 7-in-1 soil sensor, deploying Random Forest with Bayesian Optimization hyperparameter tuning to achieve 96% accuracy with a prediction time of 0.004 to 0.013 seconds. The system uses Amazon Grafana for real-time cloud dashboard visualization. Like earlier works, it classifies soil conditions for crop recommendation but does not address why conditions are unsuitable or what corrective actions should be taken to improve them.

Martelli et al. [4] conducted a three-year field trial in Italy evaluating a machine learning-based smart irrigation system for processing tomatoes. Their two-layer system used Random Forest to predict soil water potential and a genetic algorithm to generate irrigation schedules. They found that full irrigation consistently produced the highest yields, while 60% deficit irrigation caused yield losses of 61 to 78 percent. This study provides important temperature threshold information: tomato flowering is critically sensitive above 30°C, and temperatures exceeding 35°C can induce floral abortion in up to 80% of flowers. However, the system does not diagnose soil nutrient problems or recommend corrective actions to address underlying soil deficiencies.

Postolache et al. [5] designed an IoT system using ESP32 with a JXCT 7-parameter TDR sensor powered by solar energy, capable of simultaneously measuring N, P, K, pH, moisture, temperature, and electrical conductivity. They identified an important practical limitation: NPK readings from TDR-based sensors are significantly influenced by soil moisture levels, particularly after rainfall, which must be accounted for during data interpretation and preprocessing. This finding directly informs the sensor calibration and data preprocessing strategy adopted in the proposed research.

Afzal et al. [7] proposed an ensemble model combining Random Forest and XGBoost using hard voting, called RFXG, achieving 98% accuracy, precision, recall, and F1 score across 22 crop types. Their correlation analysis confirmed a strong relationship of 0.74 between potassium and phosphorus levels in soil, a finding that can inform the rule-based recommendation component of the proposed system. Notably, their study found that the Multi-Layer Perceptron achieved only 41% precision and 53.5% cross-validation accuracy, significantly underperforming compared to tree-based models such as Random Forest and XGBoost, both of which achieved 96% accuracy individually before ensemble combination.

Li et al. [8] presented an advanced IoT-XAI framework for a tomato greenhouse, combining Random Forest regression with SHAP (SHapley Additive exPlanations) analysis and Partial Dependence Plots to identify the environmental drivers of tomato fruit expansion. Their model achieved an R^2 value of 0.82 using 160 decision trees. SHAP analysis revealed that soil temperature contributed approximately 35% to the prediction, light intensity 27%, soil electrical conductivity 18%, air temperature 12%, and soil moisture 6%. This study is the most directly relevant to the proposed research in terms of applying SHAP for soil parameter explainability. However, the work focuses on greenhouse fruit expansion dynamics and does not connect SHAP outputs to actionable corrective recommendations for field farmers.

Saxena et al. [9] built a smart greenhouse platform combining ESP32-based sensor nodes with a Raspberry Pi and YOLOv8 deep learning model for tomato ripeness detection, achieving 52.8% classification accuracy, primarily limited by the small dataset of only 804 images and single-node deployment. Notably, their energy analysis confirmed that ESP32 consumes only 8.91 Wh data visualization. While the IoT architecture is relevant to the proposed work, this system does not address soil health analysis or corrective recommendations. Notably, their energy analysis confirmed that ESP32 consumes only 8.91 Wh per day, making it highly suitable for battery-powered field deployment in resource-constrained environments.

Mariammal et al. [10] proposed a Modified Recursive Feature Elimination technique for feature selection in land suitability prediction for crop cultivation. Their study found that combining soil parameters including N, P, K, zinc, copper, iron, manganese, and electrical conductivity together with environmental factors such as rainfall and season produced the best classification performance, achieving 95% accuracy using a Bagging classifier. This study confirms that careful feature selection combining both soil and environmental parameters is essential for accurate crop suitability prediction. However, the system operates entirely on a static dataset with no IoT integration, real-time monitoring capability, or corrective fertilizer recommendations for farmers.

Saha et al. [11] proposed a comprehensive IoT precision agriculture system using ESP32 with NPK, pH, moisture, and DHT11 sensors connected to the Blynk IoT cloud platform. Their Random Forest model achieved 97.35% accuracy for crop recommendation, outperforming XGBoost, LightGBM, CatBoost, Decision Tree, SVM, and KNN. Their system also incorporated fuzzy logic for irrigation scheduling, demonstrating water savings of approximately 70% compared to average-value irrigation methods. Notably, the system provides fertilizer quantity recommendations by comparing sensor readings against a standard fertilizer guide. However, like other reviewed systems, it does not assess whether current soil conditions are suitable for a specific target crop, nor does it provide any follow-up mechanism to verify whether the applied recommendations led to measurable soil improvement.

Folorunso et al. [12] developed the GeaGrow mobile application incorporating an ANN-based soil nutrient prediction pipeline combined with an expert rule-based system for fertilizer recommendations. Their system predicted NPK, organic carbon, soil texture, and pH levels from GPS coordinates alone — without requiring physical sensor deployment — achieving 97.86% accuracy for NPK prediction and 98.60% F1 score for soil texture classification. The app then applied agricultural expert rules to generate specific fertilizer quantity recommendations in kg/ha for different crops. This study is particularly relevant as it demonstrates that rule-based recommendation engines can effectively translate predicted soil analysis results into practical, actionable agricultural guidance for farmers without requiring real-time IoT sensing.

Correa-Quiroz et al. [13] developed an ESP32-based smart irrigation and climate monitoring system for greenhouses, integrating DHT11 temperature and humidity sensors, a capacitive soil moisture sensor, a GUVA-S12SD UV radiation sensor, and an HC-SR04 ultrasonic sensor for water tank level monitoring, all connected to Arduino Cloud for remote access via mobile app. The system automatically adjusts irrigation based on real-time sensor data. However, it focuses exclusively on irrigation control and does not assess soil nutrient suitability or provide crop and fertilizer recommendations.

Cheema and Pires [14] proposed an AIoT system using NodeMCU with NPK, pH, and DHT11 sensors connected to ThingSpeak cloud and Google Cloud for crop recommendation using machine learning. Among five compared algorithms, Decision Tree with AdaBoost achieved 98% accuracy while Random Forest achieved 97%. Their Pareto analysis revealed that nitrogen, phosphorus, and potassium together contribute 65.22% of the total impact on

crop recommendation decisions, while pH and rainfall add to 81.74%, confirming that NPK and pH are the most critical parameters for soil-based assessment. However, the system only recommends suitable crops and does not provide fertilizer recommendations or corrective actions — a limitation the authors themselves identify as future work.

STUDY	TECHNOLOGY	KEY CONTRIBUTION	LIMITATION / GAP
Babu et al. [1]	Arduino + Bayesian KNN	95% accuracy hybrid IoT–ML framework predicting tomato diseases (early blight, late blight, Cercospora) using NPK, temperature, humidity, and GPS sensors in Andhra Pradesh, India	Does not recommend corrective actions or monitor whether soil conditions improve over time
Islam et al. [2]	NodeMCU (ESP8266) + CatBoost + RF	Crop recommendation (CatBoost 97.5%) and fertilizer prediction (RF with GridSearchCV 99.56%) with QR-based mobile crop history tracking	No soil suitability diagnosis or feedback mechanism to verify measurable improvement in soil conditions
Tifanto et al. [3]	ESP32 + 7-in-1 sensor + RF	96% accurate crop recommendation using Random Forest with Bayesian hyperparameter tuning; Amazon Grafana real-time cloud dashboard; prediction time 0.004–0.013 s	Does not explain why soil conditions are unsuitable or provide corrective guidance
Martelli et al. [4]	μMETOS IoT + RF + Genetic Algorithm	Three-year field trial smart irrigation DSS; RF predicts soil water potential; genetic algorithm generates irrigation schedules; tomato flowering critical above 30 °C and floral abortion above 35 °C	Focuses only on irrigation scheduling; no soil nutrient diagnosis or corrective recommendations
Postolache et al. [5]	ESP32 + JXCT TDR + Solar power	Simultaneous 7-parameter soil monitoring (N, P, K, pH, moisture, temperature, EC); solar-powered system with mobile map visualization; identified moisture–NPK interference after rainfall	No ML model or crop/fertilizer recommendations; purely a monitoring system
Afzal et al. [7]	RF + XGBoost Ensemble (RFXG)	98% accuracy hard-voting ensemble across 22 crop types; strong K–P correlation (0.74); tree-	No IoT integration or real-time soil monitoring; dataset-only approach

		based models outperform MLP (41% precision)	
Li et al. [8]	IoT + RF + SHAP + PDP	Explainable AI framework: SHAP shows soil temperature 35%, light intensity 27%, soil EC 18%, air temperature 12%, soil moisture 6%; PDP identifies actionable thresholds	Applied only to greenhouse fruit expansion; SHAP explanations not linked to corrective actions
Saxena et al. [9]	ESP32 + Raspberry Pi + YOLOv8	Smart greenhouse tomato ripeness detection; ThingsBoard MQTT dashboard; ESP32 power usage only 8.91 Wh/day suitable for battery deployment	No soil nutrient analysis or fertilizer recommendations; accuracy limited by small dataset
Mariammal et al. [10]	MRFE + Bagging Classifier	Modified Recursive Feature Elimination selects N, P, K, Zn, Cu, Fe, Mn, EC + rainfall + season for crop suitability prediction	Static dataset only; no IoT integration or real-time monitoring
Saha et al. [11]	ESP32 + NPK/pH/DHT11 + Blynk + RF	97.35% accurate system integrating land mapping (NDVI/NDSI), crop recommendation, fertilizer quantity guide (kg/ha), and fuzzy irrigation saving ~70% water	Does not evaluate soil suitability for a specific crop or verify improvement after recommendations
Folorunso et al. [12]	ANN + iSDAsoil API + Flutter	GeaGrow mobile app predicts NPK, organic carbon, soil texture and pH using GPS coordinates and rule-based fertilizer guidance	Static dataset; no IoT sensing; limited to southwest Nigeria and only five crops
Correa-Quiroz et al. [13]	ESP32 + DHT11 + Capacitive + GUVA-S12SD	Smart drip irrigation system monitoring UV, soil moisture, temperature, humidity, and water level via Arduino Cloud	Focused only on irrigation control; no nutrient analysis or crop recommendation
Cheema & Pires [14]	NodeMCU + NPK/pH/DHT11 + ThingSpeak + AdaBoost	98% accurate AIoT crop recommendation; Pareto analysis shows NPK contributes 65.22% impact and pH + rainfall 81.74%	No fertilizer recommendations or corrective actions; authors state it as future work

Table 2: Summary of Reviewed Studies

1.3 Research Gap

What is Missing

The review of thirteen recent studies reveals a consistent and critical gap in the existing literature. Current IoT and machine learning systems for precision agriculture are designed to perform either prediction and classification or monitoring and visualization — but not both, and never as part of a complete decision-support cycle that guides the farmer from problem identification all the way through to verified soil improvement.

Systems such as those proposed by Babu et al. [1], Tifanto et al. [3], Afzal et al. [6], and Cheema and Pires [13] achieve high classification accuracy ranging from 95% to 98% for crop recommendation and disease prediction. However, their output terminates at a binary or categorical result. A farmer receives a label such as "not suitable" with no explanation of which specific soil parameter caused that outcome, no instruction on what corrective action to take, and no mechanism to verify whether any intervention has been effective.

Monitoring-focused systems such as Postolache et al. [5] and Saxena et al. [8] provide rich real-time sensor data but apply no diagnostic logic to the data they collect. The farmer sees numbers on a dashboard but receives no interpretation of what those numbers mean for their tomato crop or what to do about them.

Systems that do provide recommendations, such as Islam et al. [2], Saha et al. [10], and Folorunso et al. [11], generate fertilizer or crop suggestions based on soil data. However, none of these systems track whether the recommended actions have produced any measurable improvement in soil conditions after they have been applied. Most critically, no existing system tells the farmer whether their soil quality is getting better or worse over time following an intervention, and no existing system provides a revised recommendation when the first fix does not work.

Why Current Approaches are Insufficient

A complete agricultural decision-support system for tomato soil health requires four interconnected steps to be genuinely useful to a farmer. Current systems consistently address only one or two of these steps in isolation, leaving critical gaps in practical usability.

The **first step** is soil suitability assessment — determining whether current soil conditions are appropriate for tomato cultivation based on real-time sensor data collected from the field. While many reviewed systems perform this step with high accuracy, they stop here and do not proceed further [1][3][6][13].

The **second step** is parameter-level explanation using XAI — identifying which specific soil parameter is causing the unsuitability and communicating this clearly to the farmer. Without this step, a farmer who receives a "not suitable" result has no idea whether the problem is pH, nitrogen, potassium, or moisture. Li et al. [7] demonstrated that XAI technology, specifically SHAP, can identify and quantify the contribution of individual soil parameters to tomato growth outcomes. However, their system was limited to greenhouse conditions and did not extend this explanation to actionable field-level guidance for farmers.

The **third step** is intelligent corrective recommendation — providing a practical and specific corrective action based on the identified soil problem. Current recommendation systems such

as Islam et al. [2] and Folorunso et al. [11] provide static suggestions but do not connect these suggestions to a specific identified cause. No existing system uses RAG (Retrieval Augmented Generation) architecture to generate detailed, knowledge-driven, and context-specific corrective guidance for farmers based on the exact soil problem identified.

The **fourth step** is time-series feedback and verification — and this is where the most critical gap exists. No existing system monitors soil conditions over multiple time points after a corrective action has been applied. No existing system generates a clear soil quality trend showing whether soil conditions are improving or declining after a fix has been applied. No existing system provides a revised recommendation when the initial intervention is not producing sufficient improvement. Martelli et al. [4] and Saxena et al. [8] collect time-series sensor data but use it only for irrigation scheduling — not for verifying the effectiveness of soil corrections or for triggering adaptive re-recommendation.

This means that the farmer, who is the intended end user of all these systems, is left completely without support after receiving an initial recommendation. If the fix does not work, the farmer has nowhere to turn. The system gives a single output and then goes silent — leaving the farmer to figure out the rest on their own.

Justification of Novelty and Creativity

The novelty of the proposed research lies in two specific technical contributions that have not been implemented in any reviewed system. These two contributions directly address the most critical and practical gap in the existing literature.

The **first and primary novelty** is a time-series soil quality tracking module. After the system provides a corrective recommendation to the farmer, it continues to monitor soil parameter readings over multiple weeks — comparing measurements from two weeks ago, one week ago, and today. Based on this comparison, the system generates a clear and simple soil quality trend report that tells the farmer in plain terms whether their soil condition is **improving** or **declining** after the recommended fix has been applied. This gives the farmer a real, measurable answer to the most important practical question they face: *"Is what I am doing actually working?"* No reviewed system includes this component [1][2][3][4][5][6][7][8][9][10][11][12][13].

The **second and supporting novelty** is an adaptive re-recommendation mechanism that is directly triggered by the time-series tracking module. If the soil quality trend shows that conditions are not improving sufficiently after the initial recommendation has been applied, the system does not leave the farmer without guidance. Instead, it automatically retrieves updated agricultural knowledge using RAG architecture and generates a revised, context-specific recommendation based on the latest sensor readings and the observed trend. This closes the decision-support loop completely — the farmer always receives actionable guidance, whether the first intervention worked or not. No existing system in the reviewed literature includes this adaptive follow-up capability [1][2][3][4][5][6][7][8][9][10][11][12][13].

Together, these two novelties transform the proposed system from a simple prediction tool into a complete soil health management companion for tomato farmers. The system does not

just diagnose the problem — it guides the farmer through the correction process, monitors progress, and adapts its guidance until the soil reaches a suitable condition for tomato cultivation.

Evidence-Based Argument

The gap identified above is not speculative — it is directly and clearly evidenced by the limitations explicitly stated in the reviewed papers themselves.

Cheema and Pires [13] explicitly identify fertilizer recommendation and corrective guidance as future work beyond their current system. Mariammal et al. [9] acknowledge that their system lacks IoT integration and real-time monitoring capability. Li et al. [7] clearly state that their SHAP outputs are not connected to corrective recommendations for farmers. Postolache et al. [5] acknowledge that their IoT system provides no ML-based decision support whatsoever. Islam et al. [2] provide no feedback mechanism to verify whether their recommendations have led to any improvement in soil conditions. Afzal et al. [6] have no IoT or real-time sensing component at all. Folorunso et al. [11] provide static recommendations with no connection to follow-up verification or adaptive re-recommendation.

Quantitative evidence from the reviewed literature further confirms the importance and feasibility of the proposed approach. The Pareto analysis conducted by Cheema and Pires [13] confirms that nitrogen, phosphorus, potassium, and pH together account for 81.74% of the total impact on crop suitability decisions. This means that a system which tracks changes in these four parameters over time would be able to detect meaningful soil improvement or decline in the overwhelming majority of tomato soil unsuitability cases. The SHAP analysis conducted by Li et al. [7] confirmed that soil temperature contributes approximately 35%, light intensity 27%, soil electrical conductivity 18%, air temperature 12%, and soil moisture 6% to tomato growth outcomes, validating that parameter-level tracking over time using these variables would produce meaningful and reliable soil quality trend signals.

Furthermore, the finding by Postolache et al. [5] that NPK sensor readings are significantly influenced by soil moisture levels after rainfall directly justifies the need for moisture-aware data preprocessing before any time-series comparison is performed — ensuring that observed trends reflect genuine soil improvement rather than sensor artifacts caused by weather conditions.

In summary, the proposed research addresses a gap that is clearly and directly evidenced across all thirteen reviewed studies. The two novel contributions — time-series soil quality tracking and adaptive re-recommendation — have not been implemented in any existing system. They are technically justified by the convergence of real-time IoT sensing, XAI-based parameter explanation using SHAP, and RAG-based intelligent recommendation generation. They are practically motivated by the urgent real-world need of smallholder tomato farmers in Sri Lanka to receive not just a soil diagnosis, but a complete, intelligent, and verifiable course of soil correction that clearly tells them whether their soil is getting better or worse — and what to do next if it is not.

2. OBJECTIVES

2.1 Main Objective

To design and develop an IoT-based Smart Soil Health Assessment System that uses machine learning and Explainable AI to determine whether soil conditions are suitable for tomato cultivation, provide intelligent corrective recommendations, and track soil quality improvement over time through adaptive feedback monitoring.

2.2 Specific Objectives

Objective 1 — IoT Sensor System Development To design and implement a real-time soil data collection system using an ESP32 microcontroller integrated with NPK, pH, soil moisture, temperature, and humidity sensors capable of continuously monitoring soil parameters in tomato farming fields.

Objective 2 — Soil Suitability Classification To develop and train a Random Forest machine learning model that classifies real-time soil sensor data as suitable or not suitable for tomato cultivation with a target accuracy of 95% or above.

Objective 3 — XAI-Based Soil Parameter Explanation To integrate XAI technology, specifically SHAP (SHapley Additive exPlanations), into the classification model to identify and communicate which specific soil parameters are responsible for an unsuitable soil prediction and quantify their deviation from the optimal range required for tomato cultivation.

Objective 4 — RAG-Based Intelligent Recommendation To implement a RAG (Retrieval Augmented Generation) architecture that retrieves relevant agricultural knowledge from a curated knowledge base and generates specific, context-driven corrective recommendations based on the soil parameters identified as problematic by the XAI explanation layer.

Objective 5 — Time-Series Soil Quality Tracking To develop a time-series monitoring module that compares soil sensor readings across multiple time points — two weeks ago, one week ago, and today — and generates a clear soil quality trend report indicating whether soil conditions are improving or declining following the application of a recommended corrective action.

Objective 6 — Adaptive Re-Recommendation To implement an adaptive re-recommendation mechanism that automatically generates a revised corrective recommendation using the RAG architecture when the time-series tracking module detects that soil quality is not improving sufficiently after the initial recommendation has been applied, ensuring the farmer always receives actionable and up-to-date guidance.

3. METHODOLOGY

3.1 Research Approach

This research uses the **Mixed-Methods Research Methodology**. This means the research combines two types of evidence:

- **Quantitative evidence** — numbers such as model accuracy, precision, recall, and F1-score
- **Qualitative evidence** — farmer feedback on how useful and easy the system is to use

This approach is chosen because the proposed system needs to be proven in two ways. First, it must be technically accurate — the machine learning model must correctly classify soil conditions, and the time-series tracking must correctly detect whether soil is improving or declining. Second, it must be practically useful — the farmers who use the system must be able to understand the results and act on the recommendations.

The research follows three steps:

Step 1 — Build: Design and develop the complete IoT system including sensors, machine learning model, XAI explanation, RAG recommendation engine, time-series tracking, and adaptive re-recommendation.

Step 2 — Evaluate: Test every component individually and then test the complete system together using real soil data from tomato fields in Sri Lanka.

Step 3 — Validate: Compare the proposed system against existing systems from the literature to show that the proposed system solves the identified research gap.

3.2 Technical Methods and Algorithms

The proposed system has five technical components. Each one is explained below in simple terms.

Component 1 — IoT Sensor Data Collection

An **ESP32 microcontroller** will be connected to four types of sensors to collect real-time soil data from tomato farming fields:

Sensor	What it Measures	Why it is Needed
NPK Sensor (JXCT)	Nitrogen, Phosphorus, Potassium	NPK contributes 65% of soil suitability impact [13]
pH Sensor	Soil acidity and alkalinity	pH below 6.0 causes nutrient problems in tomatoes [1]
Soil Moisture Sensor	Water content in soil	Below 24% moisture suppresses tomato growth [4]
DHT11 Sensor	Temperature and humidity	Optimal tomato temperature is around 21.8°C [7]

The sensor data will be sent to the cloud using **MQTT protocol** — a simple and reliable method for sending IoT data over the internet that is widely used in agriculture [2][5].

Component 2 — Random Forest Classification Model

A **Random Forest** machine learning model will be trained to look at the soil sensor data and decide whether the soil is **suitable** or **not suitable** for growing tomatoes.

Why Random Forest? Random Forest was chosen because it consistently achieved the highest accuracy across all reviewed research papers:

Study	Accuracy
Tifanto et al. [3]	96.86%
Afzal et al. [6]	98%
Saha et al. [10]	97.35%
Cheema & Pires [13]	97%

Random Forest also provides feature importance scores which directly support the XAI explanation step.

Training:

- Dataset: Kaggle crop recommendation dataset (2,200 instances) combined with real field data from Sri Lanka
- Split: 80% for training, 20% for testing [2][13]
- Tuning: GridSearchCV will be used to find the best model settings

Component 3 — XAI Explanation Using SHAP

When the Random Forest model says the soil is **not suitable**, the farmer needs to know **why**. This is done using **SHAP (SHapley Additive exPlanations)**, which is an XAI (Explainable Artificial Intelligence) technology.

SHAP looks inside the Random Forest model and produces a simple explanation such as:

- "pH is too low — it is contributing 60% to the problem"
- "Nitrogen is too high — it is contributing 30% to the problem"

This explanation is then passed to the RAG recommendation engine to generate a specific fix for the farmer.

SHAP was selected because it was successfully used for tomato soil analysis by Li et al. [7] and is fully compatible with Random Forest models.

Component 4 — RAG-Based Recommendation Engine

After SHAP identifies the problem, a **RAG (Retrieval Augmented Generation)** system generates a specific recommendation for the farmer.

RAG works in three simple steps:

Step 1 — Store knowledge: Agricultural documents, tomato farming guides, fertilizer recommendations, and research paper findings are stored in a knowledge base.

Step 2 — Search: When SHAP says "pH is too low", RAG searches the knowledge base for the most relevant information about low pH in tomato soil.

Step 3 — Generate: A Large Language Model (LLM) uses the retrieved information to generate a clear, specific recommendation such as:

Component 5 — Time-Series Tracking and Adaptive Re-Recommendation

These are the **two novel contributions** of this research.

Novel Contribution 1 — Time-Series Soil Quality Tracking:

After the farmer applies the recommended fix, the system does not stop. It continues to monitor soil readings over time and compares:

- Readings from **2 weeks ago**
- Readings from **1 week ago**
- Readings from **today**

Based on this comparison, the system generates a simple soil quality trend report:

- **IMPROVING** — soil is getting better
- **DECLINING** — soil is getting worse
- **STABLE** — soil is not changing

This tells the farmer clearly whether the fix they applied is actually working. No existing system in the reviewed literature provides this capability [1][2][3][4][5][6][7][8][9][10][11][12][13].

Novel Contribution 2 — Adaptive Re-Recommendation:

If the time-series tracking shows that soil is **not improving**, the system automatically generates a new revised recommendation using the RAG architecture based on:

- The latest sensor readings
- The observed soil quality trend
- The previously applied recommendation

This ensures the farmer always has actionable guidance — even if the first fix did not work. No existing system in the reviewed literature includes this adaptive follow-up capability [1][2][3][4][5][6][7][8][9][10][11][12][13].

3.3 Tools and Platforms with Justification

Table 3: Tools and Platform

Tool / Platform	Purpose	Justification
ESP32	IoT microcontroller	Low cost, built-in WiFi, confirmed in multiple studies [5][8][12]
Python	ML model development	Industry standard for machine learning development
Scikit-learn	Random Forest model	Most widely used ML library, fully supports Random Forest
SHAP Library	XAI explanation	Official SHAP library, fully compatible with Scikit-learn
LangChain	RAG framework	Leading open-source framework for RAG implementation
ChromaDB	Vector database	Lightweight, open-source, easy to integrate with LangChain
Ollama / OpenAI API	LLM for recommendation	Powerful language generation for context-specific advice
MQTT Protocol	IoT data transmission	Lightweight, reliable, widely used in IoT agriculture [2][5]
Firebase	Cloud time-series storage	Real-time database for storing timestamped sensor readings
Flutter	Mobile dashboard	Cross-platform app for farmer-facing interface
Google Colab	Model training	Free GPU-enabled cloud environment for ML training

3.4 Data Collection and Ethical Considerations

3.4.1 Data Collection

Data will be collected from two sources:

Source 1 — Real-Time Field Data: Real-time soil sensor readings will be collected from tomato farming fields in Sri Lanka using the deployed ESP32 IoT sensor system. Data will be collected over a minimum of **four weeks** to provide sufficient time-series readings for the tracking module validation.

3.4.2 Ethical Considerations

Table 4: Ethical Considerations

Consideration	Action Taken
Farmer consent	Written informed consent will be obtained from all participating farmers before sensor deployment
Data privacy	All collected field data will be anonymized and stored securely with restricted access
Sensor calibration	All sensors will be calibrated before deployment to ensure accurate and reliable readings
No harm	IoT sensor deployment will not damage crops, soil structure, or farming operations

Transparency	Farmers will be fully informed about the purpose of the system and how their data will be used

3.5 Validation Strategy and Performance Metrics

3.5.1 Component Level Validation

Each component will be tested individually before the full system is tested:

Table 5: Component Level Validation

Component	How it is Tested	Target
Random Forest Model	80/20 train-test split + k-fold cross validation	Accuracy ≥ 95%
Random Forest Model	Confusion matrix analysis	Precision ≥ 90%, Recall ≥ 90%, F1 ≥ 90%
SHAP Explanation	Agricultural expert review of SHAP output	Correct parameter identification
RAG Recommendation	Agricultural expert evaluation of recommendations	Relevance ≥ 80%
Time-Series Tracking	Comparison against manual soil test results	Trend accuracy ≥ 85%
Adaptive Re-Recommendation	Expert evaluation of revised recommendations	Appropriateness ≥ 80%

3.5.2 System Level Validation

Field Testing: The complete system will be deployed in a real tomato farming field in Sri Lanka for a minimum of **four weeks**. Sensor readings, recommendations, and soil quality trends will be recorded and compared against manual soil test results to confirm the system works correctly in a real environment.

Farmer Usability Testing: A group of tomato farmers will be asked to use the system and provide feedback on:

- How clear the soil quality trend reports are
- How useful the recommendations are
- How easy the system is to understand and use

Usability will be measured using the **System Usability Scale (SUS)** with a target score of **70 or above**.

3.5.3 Performance Metrics

Metric	What it Measures	Target
Accuracy	How often the model is correct	≥ 95%
Precision	How many predicted "not suitable" are actually not suitable	≥ 90%
Recall	How many actual "not suitable" cases are correctly found	≥ 90%
F1-Score	Balance between precision and recall	≥ 90%
Soil Trend Accuracy	How correctly the system detects improving or declining soil	≥ 85%

Recommendation Relevance	How relevant the RAG recommendations are	≥ 80%
SUS Score	How easy and useful farmers find the system	≥ 70

Table 6: 3 Performance Metrics

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview

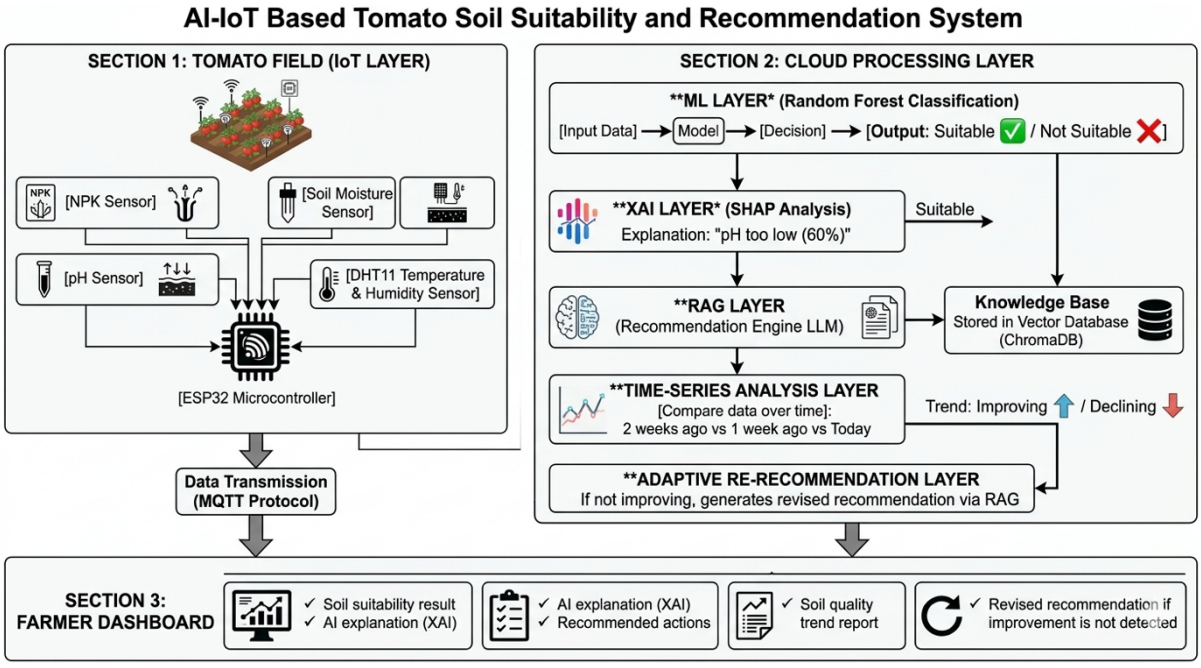
The proposed system is an **IoT-based Smart Soil Health Assessment System for Tomato Farming**. The system helps smallholder tomato farmers in Sri Lanka to:

- Check whether their soil is **suitable or not suitable** for growing tomatoes
- Understand **why** the soil is not suitable
- Receive an intelligent **corrective recommendation**
- Track whether their soil is **improving or declining** after applying the fix
- Receive a **revised recommendation** if the first fix does not work

The system has five main components:

Component	Name	Purpose
C1	IoT Sensor Layer	Collect real-time soil data from the field
C2	ML Classification Layer	Predict soil suitability for tomatoes
C3	XAI Explanation Layer	Explain which parameter is causing the problem
C4	RAG Recommendation Layer	Generate intelligent corrective recommendation
C5	Time-Series and Adaptive Layer	Track soil improvement and give revised fix if needed

4.2 System Diagram



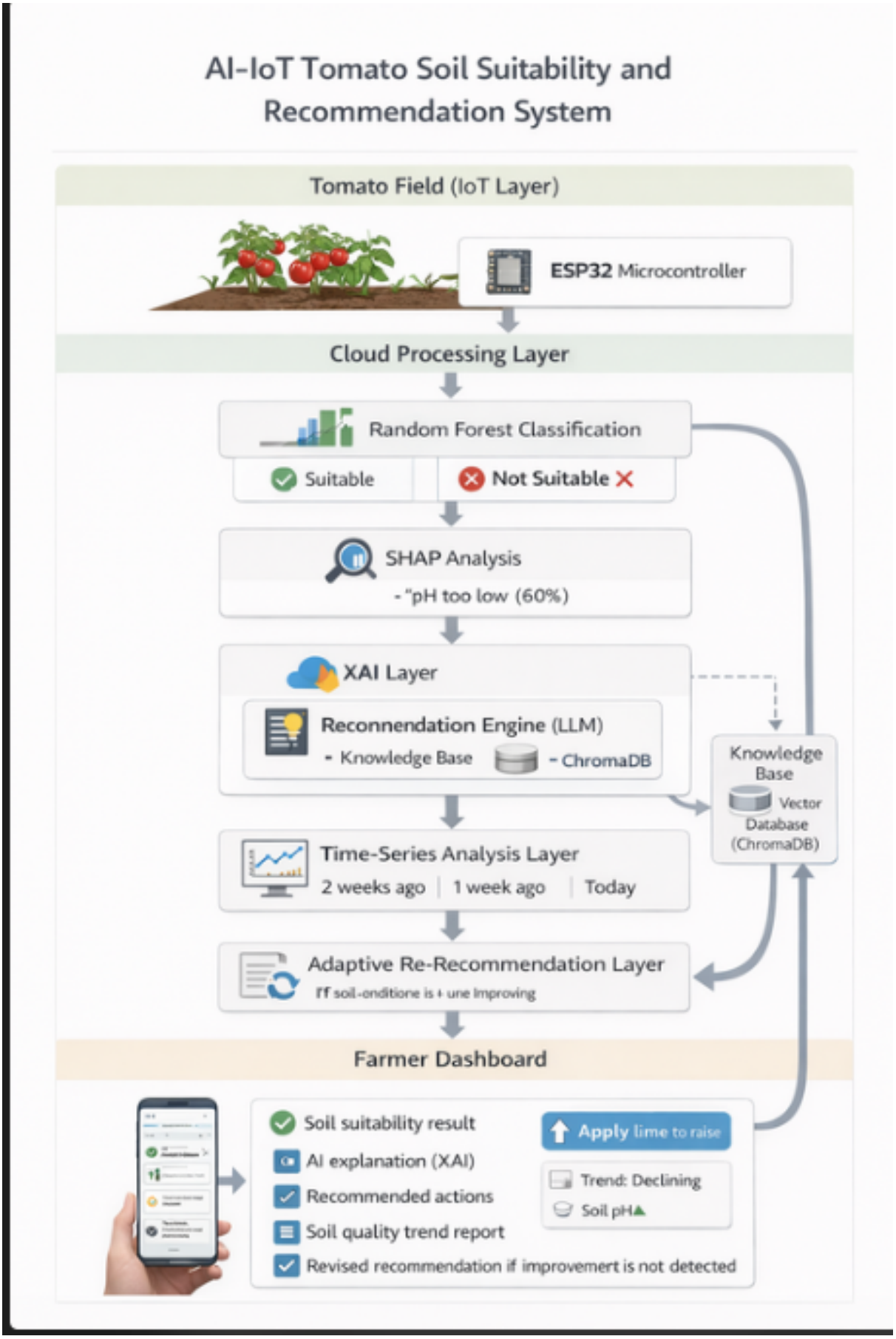


Figure 1: System Diagram

4.3 Components and Their Parts

Component 1 — IoT Sensor Layer

This component collects real-time soil data from the tomato field.

Part	Purpose
ESP32 Microcontroller	Reads all sensors and sends data to cloud
NPK Sensor (JXCT)	Measures Nitrogen, Phosphorus, Potassium
pH Sensor	Measures soil acidity and alkalinity
Soil Moisture Sensor	Measures water content in soil
DHT11 Sensor	Measures temperature and humidity
MQTT Protocol	Sends data to cloud in real time
Firebase Database	Stores all readings with timestamps

Component 2 — ML Classification Layer

This component predicts whether the soil is suitable or not suitable for tomatoes.

Part	Purpose
Preprocessing Module	Cleans and normalizes sensor data
Moisture Correction	Adjusts NPK readings for moisture influence [5]
Random Forest Model	Classifies soil as suitable or not suitable
GridSearchCV Tuner	Finds best model settings for highest accuracy

Component 3 — XAI Explanation Layer

This component explains why the soil is not suitable for tomatoes.

Part	Purpose
SHAP Explainer	Applies SHAP analysis to Random Forest prediction
Feature Contribution Calculator	Shows how much each parameter contributes to the problem
Deviation Calculator	Shows how far each parameter is from optimal tomato range
Explanation Formatter	Converts SHAP values into simple farmer-readable text

Component 4 — RAG Recommendation Layer

This component generates a specific and intelligent fix for the farmer based on the SHAP explanation.

Part	Purpose
Knowledge Base	Tomato farming guides, fertilizer documents, research findings
ChromaDB Vector Database	Stores knowledge as searchable vector embeddings
LangChain Retrieval Engine	Searches knowledge base for relevant information
LLM Generator	Generates specific recommendation using retrieved knowledge

Component 5 — Time-Series Tracking and Adaptive Re-Recommendation Layer

This is the **novel contribution** of this research. No existing system includes this component.

Table 7 : Components and Their Parts

Part	Purpose
Time-Series Data Store	Firebase stores all readings with timestamps
Trend Calculator	Compares readings from 2 weeks ago, 1 week ago, and today
Trend Report Generator	Produces improving or declining report for farmer
Improvement Threshold Checker	Checks if improvement is sufficient
Adaptive RAG Trigger	Triggers new RAG query if not improving

Trend report example:

- pH → IMPROVING
- Nitrogen → DECLINING
- Moisture → STABLE

If not improving: System automatically generates revised recommendation using RAG based on latest readings and observed trend.

4.4 Component Interactions

The five components work together in a simple step-by-step flow:

Step 1: IoT sensors collect soil data → ESP32 sends to cloud via MQTT

Step 2: Cloud receives data → preprocessing → Random Forest predicts

Step 3: If **suitable** → result sent to farmer dashboard

Step 4: If **not suitable** → SHAP analysis identifies problem parameter

Step 5: SHAP output → RAG searches knowledge base → LLM generates recommendation → sent to farmer

Step 6: Farmer applies fix → sensors continue collecting → readings stored with timestamps

Step 7: Time-series module compares 3 time points → generates trend report → sent to farmer

Step 8: If **improving** → farmer continues with fix → system keeps monitoring

Step 9: If **not improving** → adaptive RAG generates revised recommendation → sent to farmer

4.5 Data Flow

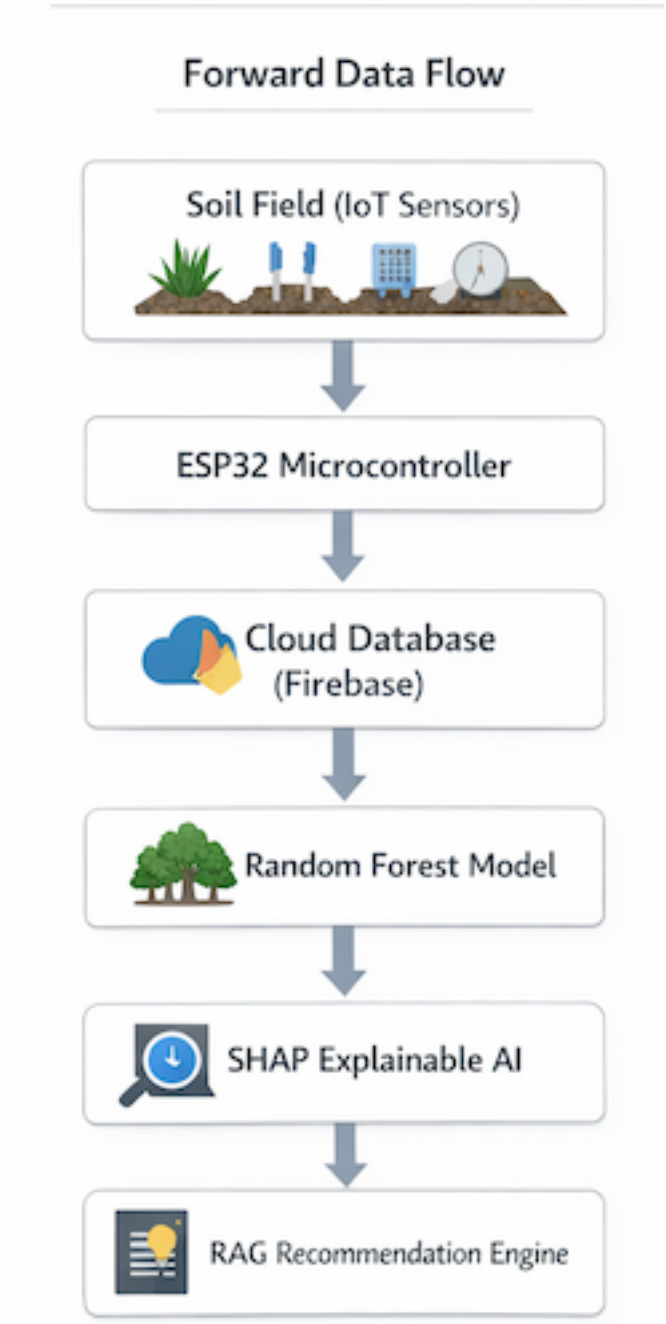


Figure 2: Data flow

4.6 Scalability and Security

Scalability:

Feature	How it Works
Multiple fields	Multiple ESP32 nodes connected to same cloud platform
Multiple farmers	Cloud and dashboard support multiple farmer accounts
More crops	Random Forest retrained with new crop data easily
Knowledge base growth	New farming documents added to RAG knowledge base anytime

Security:

Feature	How it Works
Data transmission	MQTT with TLS encryption protects data from ESP32 to cloud
Cloud storage	Firebase authentication restricts access to authorized users only
Farmer privacy	All field data anonymized and stored securely
Dashboard access	Mobile app protected with farmer login authentication

Table 8: Scalability and Security

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholder	Role
Tomato Farmer	Primary user — views soil status, recommendations, and trend reports on dashboard
Agricultural Extension Officer	Reviews and validates system recommendations
System Administrator	Manages sensors, cloud platform, and RAG knowledge base
University Supervisor	Monitors research progress and validates academic outputs

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5.2 Functional Requirements

FR01 — The system shall collect real-time soil data including N, P, K, pH, temperature, humidity, and moisture using IoT sensors connected to an ESP32 microcontroller.

FR02 — The system shall transmit all sensor data to a cloud database in real time and store each reading with a timestamp.

FR03 — The system shall classify soil as suitable or not suitable for tomato cultivation using a trained Random Forest model.

FR04 — When soil is not suitable, the system shall use SHAP analysis to identify which soil parameters are causing the problem and display this clearly on the farmer dashboard.

FR05 — The system shall generate a specific corrective recommendation using RAG architecture based on the identified soil problem.

FR06 — The system shall compare soil readings from two weeks ago, one week ago, and today to generate a trend report showing whether soil is improving, declining, or stable.

FR07 — When soil is not improving after the initial recommendation, the system shall automatically generate a revised recommendation using RAG.

FR08 — The system shall provide a mobile-accessible dashboard displaying soil results, explanations, recommendations, and trend reports in simple language.

5.3 Non-Functional Requirements

Table 9: Non-Functional Requirements

Category	Requirement
Performance	System shall generate soil prediction within 5 seconds of receiving sensor data
Performance	RAG recommendation shall be generated within 15 seconds
Accuracy	Random Forest model shall achieve minimum 95% classification accuracy
Accuracy	Model shall achieve minimum 90% precision, recall, and F1-score
Availability	System shall maintain at least 95% uptime during evaluation period
Usability	Dashboard shall be usable by a farmer with basic smartphone literacy
Usability	All outputs shall be displayed in simple English without technical jargon
Security	All farmer data shall be stored securely with restricted access

Scalability	System shall support up to 10 sensor nodes without redesign
Energy	ESP32 sensor node shall consume less than 10 Wh per day

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Primary Market: Smallholder tomato farmers in Sri Lanka, particularly in the North Central, Uva, and Central provinces where tomato cultivation is most concentrated. The target customer is a farmer who owns between 0.5 and 5 acres of tomato cultivation land and has access to a basic smartphone.

Secondary Market:

- Agricultural cooperatives managing multiple farming families
- Government agricultural extension services
- Greenhouse operators and commercial tomato farms
- Agricultural research institutions and universities

Customer Persona:

Detail	Description
Who	Smallholder tomato farmer in rural Sri Lanka
Age	25 – 55 years
Problem	Cannot afford laboratory soil tests and does not know how to fix soil problems
Need	Simple, affordable, real-time soil guidance in local language
Device	Basic Android smartphone

6.2 Value Proposition

The system provides farmers with something no existing product currently offers — not just a soil reading, but a complete answer to four critical questions:

Table 10: Value Proposition

Question	How System Answers
Is my soil good for tomatoes?	Random Forest model gives clear suitable or not suitable result
If not, why not?	SHAP XAI explanation identifies exactly which parameter is the problem
What should I do?	RAG architecture generates specific and actionable corrective recommendation
Is my fix working?	Time-series tracking shows whether soil is improving or declining week by week

This complete decision-support loop reduces crop loss caused by preventable soil problems and gives farmers the confidence to make data-driven decisions without needing expensive laboratory testing.

6.3 Revenue Model

The system will generate revenue through three streams:

Table 11: Revenue Model

Revenue Stream	Description	Price
Hardware Sale	ESP32-based IoT sensor node kit sold to farmers	LKR 15,000 per unit
Monthly Subscription	Platform subscription for dashboard access and AI recommendations	LKR 500 – LKR 1,500 per month
B2B Licensing	License the platform to agricultural extension services, NGOs, and government agencies for use with their farmer networks	Negotiated per organization

6.4 Cost Estimation

Hardware Cost per Sensor Node:

Table 12: Cost Estimation

Component	LKR	USD
ESP32 Microcontroller	1,500	$5.00
NPK Sensor (JXCT)	8,000	$26.67
Soil pH Sensor	2,500	$8.33
Soil Moisture Sensor	500	$1.67
DHT11 Temperature and Humidity Sensor	300	$1.00
Casing and Wiring	1,000	$3.33
Total Hardware Cost	**~13,800 LKR**	**~$46.00**

Platform Development Cost (One-time):

Table 13: Platform Development

Item	LKR	USD
Machine Learning Model Development	50,000	$166.67
RAG System and Knowledge Base Setup	40,000	$133.33
Mobile Dashboard Development	60,000	$200.00
Cloud Platform Setup	20,000	$66.67
Total Development Cost	**~170,000 LKR**	**~$566.67**

6.5 Pricing Strategy

The pricing strategy is designed to be affordable for smallholder farmers in Sri Lanka while ensuring long-term platform sustainability:

Table 14: Pricing Strategy

Tier	Target	LKR	USD
Basic	Individual farmer — sensor kit only	LKR 15,000 one-time	~$50.00 one-time
Standard	Individual farmer — sensor kit + dashboard subscription	LKR 15,000 + LKR 500/month	~$50.00 + $1.67/month
Premium	Individual farmer — sensor kit + full AI features + RAG recommendations	LKR 15,000 + LKR 1,500/month	~$50.00 + $5.00/month
B2B	Agricultural organizations and government agencies	Custom licensing fee	Custom licensing fee

The monthly subscription cost of LKR 500–1,500 is intentionally set below the average daily wage of a Sri Lankan farmer to ensure affordability and encourage adoption.

6.6 Competitive Advantage

The proposed system has four key competitive advantages over any existing product:

Table 15: Competitive Advantage

Advantage	Description
Complete decision-support loop	The only known system combining IoT sensing, XAI diagnosis, RAG recommendation, and time-series tracking in one platform
Explainable AI	Unlike other systems that give only a yes or no result, this system explains exactly why the soil is unsuitable using SHAP
Adaptive recommendations	Unlike static recommendation tools, this system generates revised recommendations if the first fix does not work
Low cost	Designed specifically for smallholder farmers in developing countries with affordable hardware and subscription pricing

Existing commercial tools such as general crop recommendation apps and standalone soil sensors do not provide the complete decision-support loop described above. No reviewed

research system combines all four components into a single unified platform [1][2][3][4][5][7][8][9][10][11][12][13].

6.7 Intellectual Property Considerations

The novel combination of the following four components constitutes an original system design:

1. Real-time IoT soil data collection using ESP32 sensors
2. XAI-based soil suitability explanation using SHAP
3. RAG architecture for intelligent corrective recommendation generation
4. Time-series soil quality tracking with adaptive re-recommendation

This original design can be protected through the following intellectual property mechanisms under Sri Lankan IP law:

Table 16: Property Considerations

IP Type	Application
Software Patent	The novel combination of SHAP diagnosis, RAG recommendation, and time-series adaptive feedback can be filed as a software patent
Copyright	The system source code, dashboard design, and knowledge base content are protected as original intellectual works
Trademark	The system name and brand can be registered as a trademark for commercial protection

7. BUDGET AND JUSTIFICATION

7.1 Hardware Costs

Table 17: Budget and Justification

Item	Qty	Unit Cost (LKR)	Total (LKR)	Total (USD)	Justification
ESP32 Microcontroller	2	1,500	3,000	~$10.00	Primary data acquisition unit — low cost, built-in WiFi, confirmed in multiple studies [5][8][12]
NPK Soil Sensor (JXCT RS485)	1	8,000	8,000	~$26.67	Measures N, P, K simultaneously — compatible with ESP32 [2][5]
Soil pH Sensor	1	2,500	2,500	~$8.33	Essential for tomato soil suitability assessment [1][7]
Soil Moisture Sensor	1	500	500	~$1.67	Required to monitor soil water content [4]
DHT11 Temperature and Humidity Sensor	1	300	300	~$1.00	Low cost and accurate — widely used in IoT agriculture [8][12]
Jumper Wires, Breadboard, Power Supply	—	—	1,500	~$5.00	Required for prototyping and hardware assembly
Casing and Weatherproofing	1	1,000	1,000	~$3.33	Protects hardware during field deployment
Hardware Subtotal			**16,800 LKR**	**~$56.00**	

7.2 Software Licenses

Item	Cost	Justification
Python	Free	Open source — industry standard for ML development
Scikit-learn	Free	Open source ML library for Random Forest model
SHAP Library	Free	Open source XAI library compatible with Scikit-learn
LangChain	Free	Open source RAG framework for recommendation engine
ChromaDB	Free	Open source vector database for RAG knowledge base
Flutter	Free	Open source mobile framework for farmer dashboard
Software Subtotal	**LKR 0 / $0**	All required software tools are open source and free

7.3 Cloud Services

Item	Cost	Justification
Firebase	Free	Free tier sufficient for prototype — real-time database for sensor data storage and time-series tracking
Google Colab	Free	Free GPU-enabled environment for Random Forest model training
Ollama	Free	Free local LLM for RAG recommendation generation without API costs
Cloud Subtotal	**LKR 0 / $0**	All required cloud services are available on free tier for prototype stage

7.4 Contingency

Item	LKR	USD	Justification
Miscellaneous and Contingency (10%)	1,680	~$5.60	Buffer for unexpected hardware replacement, additional materials, or travel costs during field testing

7.5 Total Budget Summary

Category	LKR	USD
Hardware Costs	16,800	~$56.00
Software Licenses	0	$0.00
Cloud Services	0	$0.00
Contingency	1,680	~$5.60
TOTAL	**~18,480 LKR**	**~$61.60**

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Overview

This project is completed by a single student researcher over seven months from March to October. The work is divided into four stages based on the campus submission and presentation deadlines.

Table 18: Work Breakdown Structure

Stage	Period	Deadline	Target
Stage 1	March 1 – 15	March 15	Proposal submission
Stage 2	March 16 – May 3	May 4 – 6	50% complete — individual components working
Stage 3	May 7 – August 16	August 17 – 19	100% complete — fully integrated system
Stage 4	August 20 – October 11	October 12 – 14	Final product ready to present

8.2 Task Breakdown Per Member

Task No	Task	Member	Stage	Period
T01	Literature review completion	Student	Stage 1	March 1 – 10
T02	System architecture design	Student	Stage 1	March 1 – 12
T03	Sensor selection and hardware procurement	Student	Stage 1	March 10 – 15
T04	ESP32 sensor node assembly and testing	Student	Stage 2	March 16 – April 5
T05	Cloud platform setup and MQTT integration	Student	Stage 2	March 16 – April 10
T06	Dataset preparation and preprocessing	Student	Stage 2	April 1 – 15
T07	Random Forest model training and evaluation	Student	Stage 2	April 10 – 25
T08	SHAP integration and XAI testing	Student	Stage 2	April 20 – May 3
T09	RAG system and knowledge base development	Student	Stage 3	May 7 – June 15
T10	Time-series tracking module development	Student	Stage 3	June 1 – July 1
T11	Adaptive re-recommendation module development	Student	Stage 3	July 1 – July 20
T12	Dashboard development and integration	Student	Stage 3	July 15 – August 10
T13	System testing with real soil data	Student	Stage 3	August 1 – 15
T14	Full system performance evaluation and tuning	Student	Stage 4	August 20 – September 15

T15	Farmer usability testing	Student	Stage 4	September 1 – 30
T16	Final report writing and submission	Student	Stage 4	September 15 – October 10

8.3 Timeline Alignment

Stage 1 — March 1 to 15

Target: Proposal Submission — March 15

Task	Period
T01 — Literature review completion	March 1 – 10
T02 — System architecture design	March 1 – 12
T03 — Sensor selection and hardware procurement	March 10 – 15

Must be ready by March 15:

- Full proposal report completed and submitted

Stage 2 — March 16 to May 3

Target: Progress Presentation 1 — May 4, 5, 6 — 50% Complete

Task	Period
T04 — ESP32 sensor node assembly and testing	March 16 – April 5
T05 — Cloud platform setup and MQTT integration	March 16 – April 10
T06 — Dataset preparation and preprocessing	April 1 – 15
T07 — Random Forest model training and evaluation	April 10 – 25
T08 — SHAP integration and XAI testing	April 20 – May 3

Must be ready by May 4:

- ESP32 collecting real soil data from field
- Sensor data sending to cloud via MQTT
- Random Forest model working with minimum 95% accuracy
- SHAP producing correct soil parameter explanations
- Each component demonstrated individually

Stage 3 — May 7 to August 16

Target: Progress Presentation 2 — August 17, 18, 19 — 100% Complete

Task	Period
T09 — RAG system and knowledge base development	May 7 – June 15
T10 — Time-series tracking module development	June 1 – July 1
T11 — Adaptive re-recommendation module development	July 1 – July 20
T12 — Dashboard development and integration	July 15 – August 10
T13 — System testing with real soil data	August 1 – 15

Must be ready by August 17:

- RAG system generating intelligent corrective recommendations
- Time-series tracking module showing soil improving or declining trends
- Adaptive re-recommendation working when soil is not improving
- Farmer dashboard displaying all outputs clearly
- All components fully integrated with teammate components

Stage 4 — August 20 to October 11

Target: Final Submission — October 12, 13, 14

Task	Period
T14 — Full system performance evaluation and tuning	August 20 – September 15
T15 — Farmer usability testing	September 1 – 30
T16 — Final report writing and submission	September 15 – October 10

Must be ready by October 12:

- Complete system fully tested with real soil data
- All performance metrics achieved
- Farmer usability testing completed with SUS score
- Final report completed and submitted
- System polished and ready to present to real users

8.4 Balanced Workload Distribution

Stage	Period	Number of Tasks	Percentage of Total Work
Stage 1 — Research and Design	March 1 – 15	3 tasks	10%
Stage 2 — Individual Components	March 16 – May 3	5 tasks	30%
Stage 3 — Full Integration	May 7 – August 16	5 tasks	40%
Stage 4 — Final Product	August 20 – October 11	3 tasks	20%
Total	**7 months**	**16 tasks**	**100%**

8.5 Key Milestones

Table 19: Key Milestones

Milestone	Target Date
Proposal report submitted	March 15
IoT hardware assembled and cloud connected	April 10
Random Forest model trained with SHAP integrated	May 3
Progress Presentation 1 — 50% complete	May 4 – 6
RAG system and knowledge base completed	June 15
Time-series and adaptive modules completed	July 20
Full system integrated and dashboard ready	August 15
Progress Presentation 2 — 100% complete	August 17 – 19
Performance and usability testing completed	September 30
Final report submitted	October 10
Final product presentation	October 12 – 14

9. GANTT CHART

The following table represents the semester-wise project timeline with key milestones.

Table 20: Gantt Chart

Task	Mar 1–15	Mar 16–31	Apr 1–30	May 1–6	May 7–31	Jun 1–30	Jul 1–31	Aug 1–19	Aug 20–31	Sep 1–30	Oct 1–14
Literature Review	■										
System Architecture Design	■										
Hardware Procurement	■										
ESP32 Assembly and Testing		■	■								
Cloud and MQTT Setup		■	■								
Dataset Preparation			■								
Random Forest Model Training			■								
SHAP Integration and Testing			■	■							
RAG System Development					■	■					
Time-Series Module Development						■	■				
Adaptive Re-Recommendation							■				
Dashboard Development							■	■			
System Testing								■			
Performance Evaluation									■	■	
Farmer Usability Testing										■	
Final Report Writing									■	■	■

Key Milestones:

- Proposal submitted — **March 15**
- Progress Presentation 1 — 50% complete — **May 4 – 6**
- Progress Presentation 2 — 100% complete — **August 17 – 19**
- Final product presentation — **October 12 – 14**

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11. APPENDICES

Appendix A — Survey Instruments

A.1 Survey Overview

A structured survey will be conducted with **Agricultural Inspectors** and **tomato farmers** in Sri Lanka to validate the system requirements and gather practical insights about soil health challenges faced in the field.

The following steps will be taken to conduct the survey:

- Agricultural Inspectors have been contacted and have agreed to provide guidance
- A meeting with Agricultural Inspectors is scheduled for next month
- Direct field visits to tomato farming areas will be conducted to survey farmers
- Survey responses will be used to validate system requirements and improve recommendation accuracy

A.2 Survey Questionnaire — Agricultural Inspectors

Table 21: Survey Questionnaire - Agricultural Inspectors

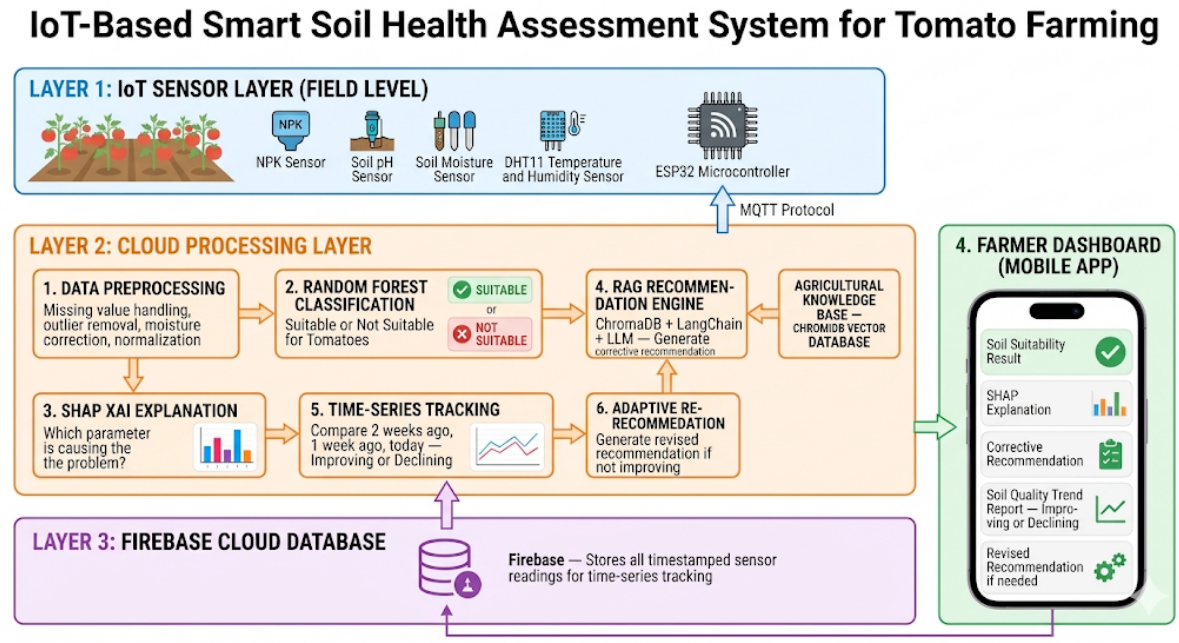
No	Question
1	What are the most common soil problems you observe in tomato farming fields in Sri Lanka?
2	Which soil parameters do you check most frequently when visiting tomato farms?
3	What corrective actions do you most commonly recommend to farmers for soil problems?
4	How long does it typically take for a farmer to see improvement after applying a soil correction?
5	Do farmers currently have access to any digital tools for soil health monitoring?
6	Would a mobile system that tells farmers whether their soil is improving or declining be useful?
7	What information would you want to see on a farmer-facing soil health dashboard?
8	What challenges do farmers face when trying to correct soil problems?

A.3 Survey Questionnaire — Tomato Farmers

Table 22 : Survey Questionnaire - Tomato Farmers

No	Question
1	How many acres of land do you use for tomato cultivation?
2	Have you ever experienced poor tomato yield due to soil problems?
3	How do you currently check whether your soil is suitable for tomatoes?
4	Do you use any digital tools or mobile apps to help with farming decisions?
5	Would you be willing to use a low-cost IoT sensor system to monitor your soil?
6	Would you find it helpful if the system told you exactly which soil parameter is causing a problem?
7	Would you find it helpful if the system tracked whether your soil is getting better or worse over time?
8	How much would you be willing to pay per month for such a system?

Appendix B — Detailed Diagrams



Appendix C — Preliminary UI Mockups

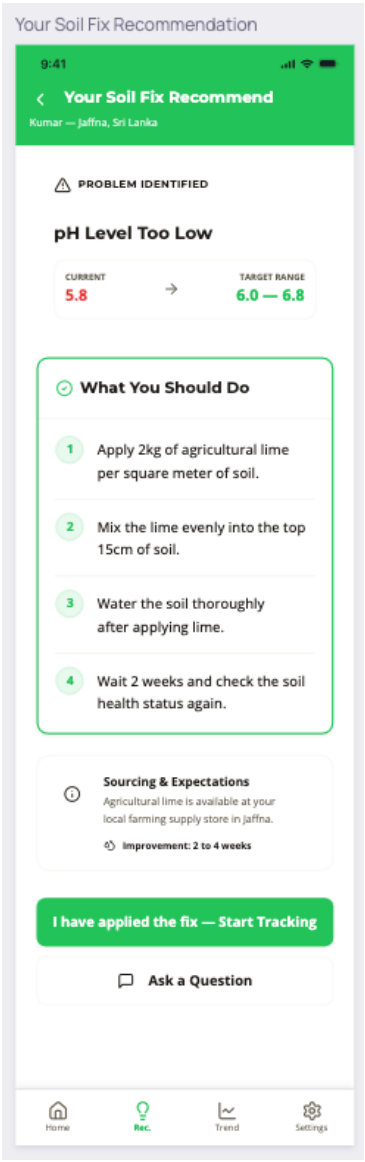


Figure 5: Mobile Dashboard - Main Screen

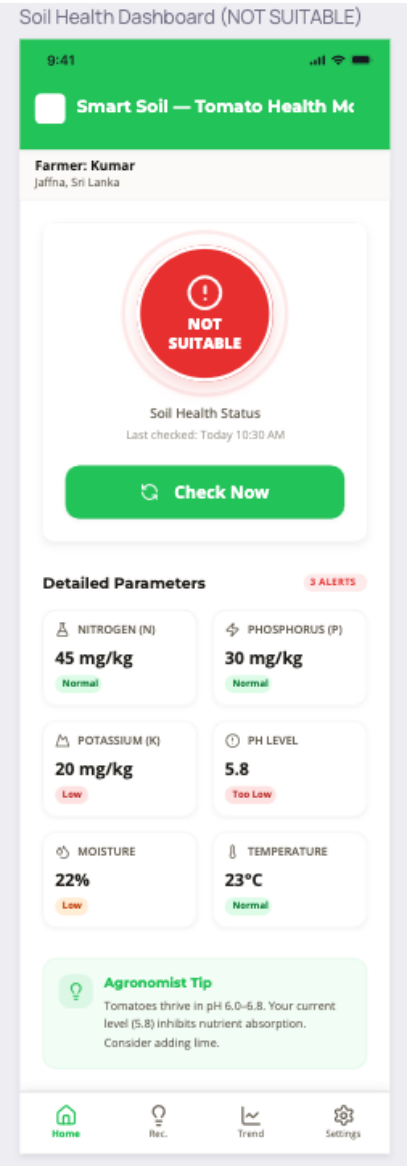


Figure 4: Mobile Dashboard - SHAP Explanation Screen

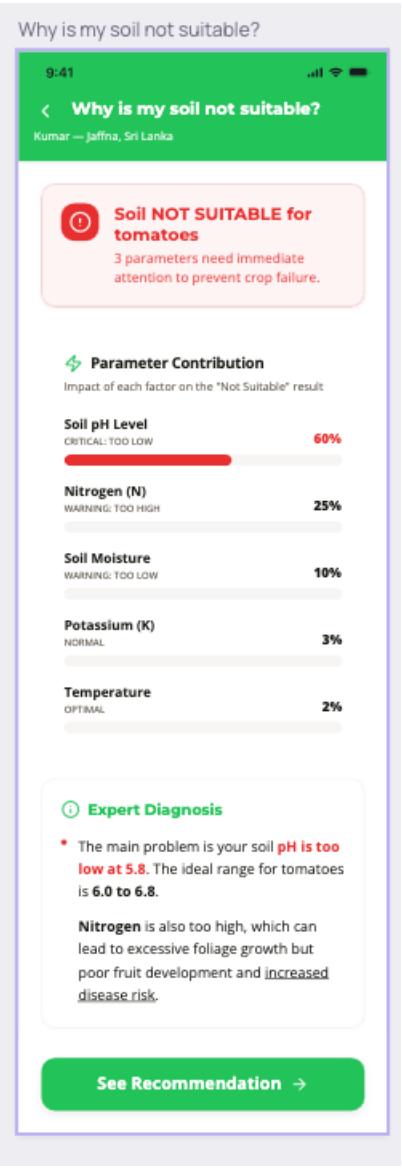


Figure 3 :Mobile Dashboard - RAG Recommendation Screen

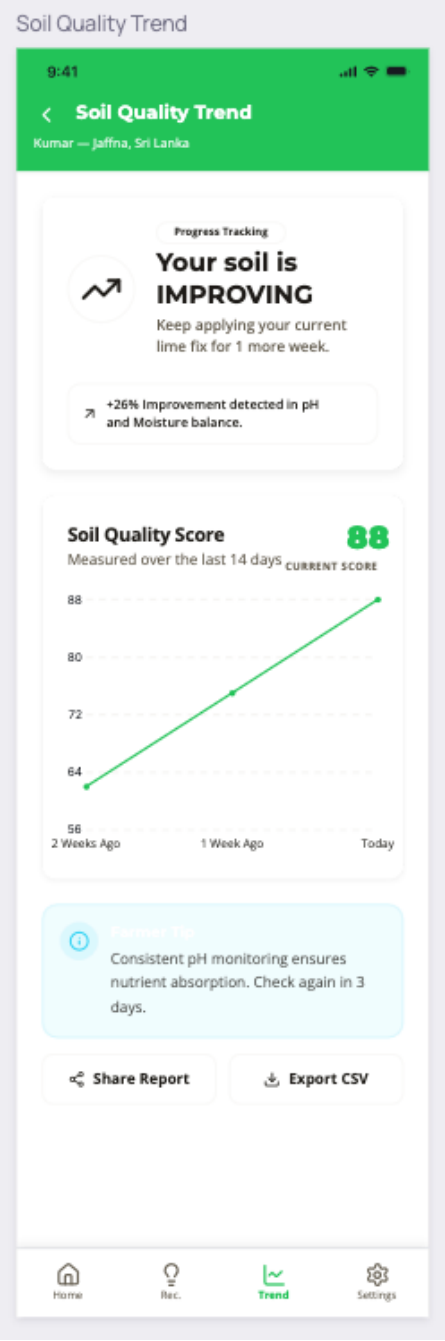


Figure 7 :Mobile Dashboard - Time-Series Trend Screen

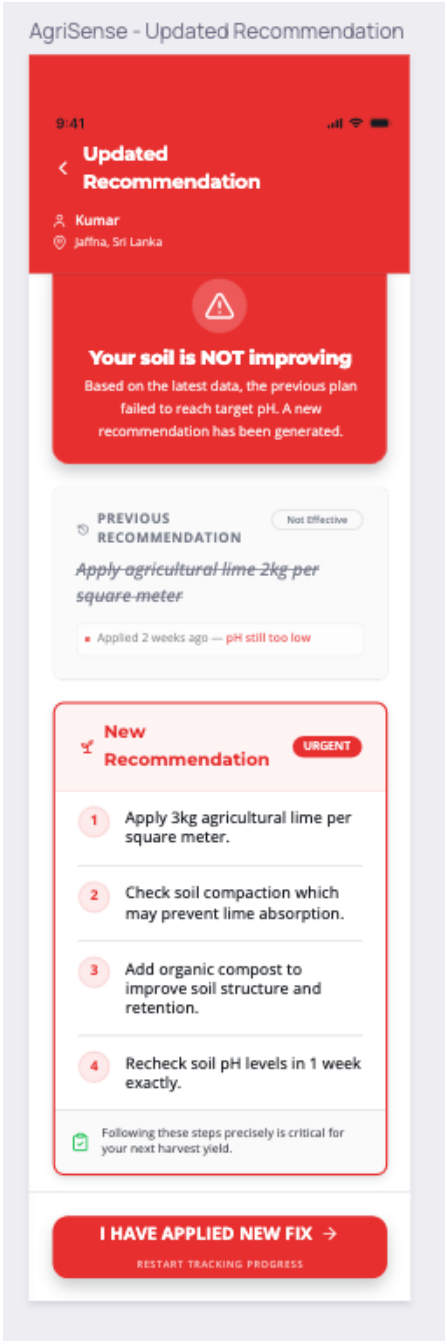


Figure 6 :Mobile Dashboard - Adaptive Re-Recommendation Screen

Appendix D — Risk Analysis

D.1 Risk Register

Table 23 :Risk Register

No	Risk	Likelihood	Impact	Mitigation
R01	NPK sensor readings inaccurate due to soil moisture after rainfall	High	High	Apply moisture correction preprocessing as identified by Postolache et al. [5]
R02	Random Forest model accuracy below 95% target	Medium	High	Use GridSearchCV hyperparameter tuning and collect additional field data
R03	RAG system generates irrelevant recommendations	Medium	High	Curate high quality knowledge base and validate recommendations with Agricultural Inspector
R04	ESP32 hardware failure during field deployment	Medium	Medium	Purchase two ESP32 units and keep one as backup
R05	Insufficient real field data collected for time-series validation	Medium	High	Begin field data collection early and use generated synthetic data as supplement
R06	Firebase free tier storage limit exceeded	Low	Medium	Monitor storage usage and upgrade to paid tier if necessary
R07	Farmers unable to understand dashboard outputs	Medium	High	Conduct usability testing early and simplify dashboard language based on feedback
R08	LLM generates incorrect agricultural advice	Medium	High	Validate all RAG outputs against verified agricultural guidelines before deployment
R09	Project timeline delayed due to hardware procurement	Low	Medium	Procure all hardware components immediately after proposal approval
R10	Integration with teammate components fails	Medium	High	Plan integration sessions early and maintain clear API documentation

D.2 Risk Summary

Risk Level	Number of Risks
High Impact	7 risks
Medium Impact	3 risks
Low Impact	0 risks

Appendix E — Ethical Considerations

E.1 Research Ethics Statement

This research involves the collection of real-time soil data from tomato farming fields in Sri Lanka and direct interaction with farmers and Agricultural Inspectors through surveys and usability testing. The following ethical principles will be followed throughout the research:

Table 24: Ethical Considerations

Principle	Action
Informed Consent	Written consent will be obtained from all farmers and Agricultural Inspectors before any data collection or system deployment
Voluntary Participation	All participants have the right to withdraw from the study at any time without any consequence
Data Privacy	All personal information collected from participants will be kept confidential and will not be shared with any third party
Data Anonymization	All field data and survey responses will be anonymized before use in research analysis
No Harm	IoT sensor deployment will not damage crops, soil structure, or farming operations in any way
Transparency	All participants will be fully informed about the purpose of the research and how their data will be used
Data Security	All collected data will be stored securely in a password-protected cloud database with restricted access
Academic Integrity	All research findings will be reported honestly and accurately without any fabrication or misrepresentation